

Git & GitHub Fundamentals Report

Git

Git is a distributed version control system that tracks changes in source code, allows multiple developers to collaborate, maintains history, and enables reverting to previous versions.

GitHub

GitHub is a cloud-based platform for hosting Git repositories. It provides collaboration features such as issues, pull requests, code reviews, and CI/CD integration.

Git Workflow

A Git workflow is a set of rules and practices that define how a development team collaborates using Git, determining how changes are branched, merged, and delivered across the software lifecycle. It serves as a standardized recipe for managing source code, ensuring that teams work consistently to avoid merge conflicts and maintain a clean, stable production history.

Branching & Merge

A branch is an independent line of development. Developers create feature branches to isolate work. After testing, branches are merged into the target branch. Merges may be fast-forward or create a merge commit.

Rebase

Rebase moves or reapplies commits onto another branch, creating a cleaner linear history. Unlike merge, it rewrites commit history, so it should be used carefully on shared branches.

Pull Requests

A Pull Request (PR) proposes changes from one branch to another. Team members review code, discuss improvements, run automated checks, and approve the changes before merging.

README.md

README.md is the first document visitors read. It typically contains the project overview, features, installation steps, usage instructions, folder structure, technologies used, contribution guidelines, and license.

.gitignore

.gitignore lists files and folders that Git should ignore, such as build artifacts, log files, IDE settings, dependencies (node_modules), compiled files, and sensitive files like .env.

Conclusion

Git and GitHub simplify version control and collaboration. By using branches, pull requests, README.md, and .gitignore effectively, developers can maintain organized, secure, and collaborative software.